

Integrating Security and Privacy Goals: Sustainable and Ethical Data Governance in Migration Technologies

A critical analysis of the 2024 Greek Ministry of Migration ruling and EU-US Data Partnership within the framework of Global Sustainability, SGD 10 and SGD 16.

EXECUTIVE SUMMARY

At the end of April 2025, the United Nations reported a new record-breaking figure that 122 million people have been forcibly displaced due to violence, persecution, or conflict. In today's digital age, use of biometric data, surveillance and predictive technologies regarding the migration of vulnerable populations, such as refugees and asylum seekers, is becoming the norm globally used by governments, cooperations, and nongovernmental organizations (NGOs). While there are numerous benefits such as greater security at borders, scalability, digitization and cost efficiency, the ethical oversight and governance frameworks have not evolved efficiently.

This has raised ethical, legal, and humanitarian concerns such as data privacy and protection, consent, limited accountability, weak transparency, and risks of misrepresentation. Such failures can threaten the progress of the two of the United Nations Sustainable Development Goals (SDGs): Goal 16 - Peace, Justice and Strong Institutions which promote inclusive and peaceful societies and Goal 10 - Reduced Inequalities which aims to reduce economic and social disparities within and among countries. The historic landmark penalty imposed upon by Hellenic Data Protection Authority in Greece on the Ministry of Migration and Asylum has shone a light on the significant GDPR violations within a public body exposing significant gaps in data governance and accountability. In addition, the recent Enhanced Border Security Partnership (EBSP) between the EU and U.S. agencies, which would grant access to real time biometric queries on EU systems by law enforcement authorities, has been criticized by privacy advocates with regards to the potential risks. This also touches on SGD 17, which aims to provide greater cross border cooperation to achieve global sustainable development through inclusive partnerships.

This report examines the challenges involved in this evolving landscape by considering existing governance, regulatory frameworks, privacy regimes, and the ethical, legal and social implications by drawing lessons learned from real-world case studies. Robust solutions are then proposed that underscore the necessity of strong data governance to ensure fair, transparent, and sustainable systems that also protect both the data and the fundamental rights of vulnerable and marginalized populations.

Key findings show:

- The ethical concerns of the European Union method of governing the data on migration are identified at a significant level. Regardless of the effective legal frameworks such as the General Data Protection Regulation (GDPR), the ethical implementation is still weak. The growing reliance on biometric and algorithmic systems, and several large databases, such as the European Dactyloscopy System (EURODAC), Schengen Information System (SIS), Entry/Exit System (EES), to name a few, has increased the concern about respecting transparency, accountability, and protection of vulnerable populations. Lack of effective oversight mechanisms and lack of access to justice are some of the obstacles to the implementation of SDGs 10 and 16 that propose equality and well-established institutions. As a result, to achieve long-term sustainability, the EU needs to strengthen human-centered governance, fair data protection, and institutional confidence in the migration management systems.
- Data privacy and protection in EU migration management should be at the

forefront regarding biometric data collection and cross-border sharing. Despite the GDPR's strong legal framework, the enforcement of the fundamental principles remains inconsistent. This has resulted in personal data and surveillance technologies being used for basic human rights being threatened and can result in deportations or asylum denial. Therefore, it is important that privacy principles are integrated into systems from the development stage and there is independent oversight. In addition, transparent data governance is essential to uphold accountability and human rights. By ensuring data protection is upheld, it provides legal compliance but most importantly the dignity and safety of vulnerable populations.

- The ethical, legal and social considerations are crucial in data driven technologies with the use of biometric data being used to improve efficiency and fraud prevention they raise concerns regarding data privacy, consent, bias, and data breaches that must be addressed. The different ethical theories studied highlight the importance of balancing the utilitarian views with fairness, equality and respect for human dignity While biometric systems may not consider cultural, religious or personal values, this can result in social alienation resulting in further bias, discrimination, and inequality. Legally, cross-border data sharing can be quite complex and inconsistent regarding differing international law policies and can threaten transparency and institutional accountability.
- To ensure the principles of SDG 10 (reducing inequality) and SDG 16 (building strong institutions), these migration systems need to do more than just follow laws like the GDPR and EU AI Act. They must actively ensure fairness, transparency, and respect for human rights. This means holding authorities accountable, regularly reviewing privacy and ethics, involving migrants, NGOs, and

civil society in decision-making. Strong data protection techniques such as encryption, anonymization, and checks for AI bias are crucial. Additionally, clear consent procedures, minimal data collection, and accessible channels for people involved to raise concerns should all be considered. Oversight can be ensured through independent audits, open reporting, and proper staff training. By this comprehensive approach, the EU can make its migration systems not only more secure and efficient, but also fairer, ethical, and earn public trust.

There is a critical need for a strong robust policy-driven framework that captures the complex realities of the ongoing challenges regarding safeguarding privacy, receiving informed consent, bureaucratic harms, and protecting data of the actors involved. The recommendations of this study aim to provide actionable and sustainable frameworks for policy makers, governments and humanitarian organizations to ensure that basic human rights, accountability and transparency and embedded into the heart of these technologies. Until the dignity and fundamental rights of people are established, then only can we truly value the use of these sociotechnical systems.

I. GOVERNANCE AND REGULATORY FRAMEWORKS FOR SUSTAINABLE CHALLENGE

EU migration management is one of the most challenging and ethically debatable fields of digital data governance and regulation. In this respect, data is not only seen as a technical or administrative tool but can be used as a weapon from a political standpoint as it can be used to determine basic human rights and services access. Sensitive biometric and personal information, including fingerprints, facial recognition, surveillance and behavioural analytics, are now being utilized to control migration flows, identify individuals, and provide security. Significant ethical and governance issues arise from the reliance on complex data systems that extend beyond concerns of technical effectiveness. In this section, the critical review of the governance and regulatory frameworks aimed at controlling such practices is conducted at every stage of the data

process, and the effects on SDG 10 (Reduced Inequalities) and SDG 16 (Peace, Justice, and Strong Institutions) are discussed [1].

The General Data Protection Regulation (GDPR) is at the centre of the European Union data governance mechanism and one of the most detailed privacy and data protection laws worldwide [2]. The GDPR lays out the principles of lawful data processing: the limitation of purpose, minimization of data, informed consent, and accountability. The complement to the GDPR are other laws, like the EU Charter of Fundamental Rights, which formally qualifies the right to privacy and data protection as a primary human right, the UK Data Protection Act of 2018 and the new EU Artificial Intelligence (AI) Act that further protects high-risk technologies, including those employed in migration management [3]. All these frameworks are aimed at making data use in accordance with ethical principles, transparency, and social responsibility.

Despite the continuing failures, the problem does not seem to be the lack of regulatory frameworks, but the difficulty of putting governance into practice. The audit of a few EU migration systems has revealed the lack of integrity in the form of missing Data Protection Impact Assessments (DPIAs), lack of effective implementation of oversight decisions, and lack of coordination between different levels of government. These results show how human rights protection policies can easily fail at the implementation level. Socio-Technical Systems (STS) that is comprised of technology, people, procedures, and social considerations, in which data processing is embedded, still exhibit structural and moral flaws [4].

It is also alarming that accountability and risk management has failed in many governance structures. The Greek case on data protection is a clear example of this. The Ministry of Migration and Asylum was fined based on running surveillance systems, Kentauros (Centaur) and Hyperion, without carrying out proper Data Protection Impact Assessments (DPIAs) [6]. The DPIA is regarded as a significant part of accountable data governance to ensure that processing of substantial risk is protected, reasonable, and morally appropriate. It represents a systemic ineffectiveness of the Data Governance

Council and the data stewards, who failed to recognize and avert any possible risks from this massive collection of biometric and surveillance data. This irresponsibility prioritizes operational efficiency over ethical considerations leading to considerable fines from regulatory authorities and further widening greater mistrust within communities.

Another ethical concern is the violation of the principles of limited purpose and minimization of data. Biometric and personal data that has initially been collected for humanitarian or administrative contexts, have been reused in secondary applications including sharing with foreign security agencies to aid in the facilitation of deportation processes [7]. This tension is exemplified by the most recent proposals through the Enhanced Border Security Partnership (EBSP), who have proposed access by the U.S. authorities to core EU police and migration databases, including the European Dactyloscopy System (EURODAC), the Schengen Information System (SIS), the Visa Information System (VIS), the Entry/Exit System (EES), the European Travel Information and Authorization System (ETIAS) and the European Criminal Records Information System of Third Country Nationals (ECRIS-TCN) through biometric and biographic queries.

On one hand, supporters claim that these types of integrations might improve trans-Atlantic security collaborations. On the other hand, opponents warn that it might undermine the values or principles of the GDPR (specifically touching on Chapter V), reduce transparency, and reduce access to corrective or deletion measures. This is especially worrisome for migrants and asylum seekers who may face struggles to challenge system bias or unclear automated judgment [8]. Such repurposing compromises data security and integrity since it puts individuals at risk. This is further worsened by the absence of thorough data provenance or the capability to trace the source of data, processing of data, and by whom. As soon as the data is no longer in its source of collection, refugees and migrants do not have any control and monitoring over how it is used, significantly limiting their chances to contradict falsehoods or misuse it.

Governance structures that do not rely on technical compliance and are not regulated or enforced by humans and values cannot adequately protect

rights. The result is a cycle of repeat contraventions of ethics that destroy public confidence and damage the sustainability goals that these systems are supposed to reinforce. These systemic weaknesses lie in the organizational tier of the STS, in which high-level data governance policies are poorly enforced. They are additionally supported by the societal layer, where political debates on migration are mostly dominated, which generally tends to further justify the use of intrusive data for national security [9]. In this kind of environment, ethical risks and the degradation of privacy and data security embedded into these technologies, are becoming the norm, unfortunately targeting vulnerable groups.

To make data governance compatible with sustainability goals, a more initiative-taking, rights-based model is needed, one that instils ethics at each step of the data management process. This would go beyond compliance with procedures and include the implementation of Ethical Impact Assessments (EIAs) in conjunction with the conventional DPIAs [10]. Whereas DPIAs are mostly related to risk and legality, EIAs are associated with a deeper level of ethical questions, including: Does this system uphold human dignity? Is it reasonable and fair? Is it beneficial to the common good? Such evaluations implemented during the design phase would make sure that before systems are rolled out, ethical reflection makes a difference in technical and policy making

In addition, data provenance and stewardship should be improved ensuring accountability. All the phases of the data lifecycle, the collection through to the disposal stage, must be well documented which includes who accessed the data, the reason and under what authorization [11]. Governmental and humanitarian organizations may put in place formal data custodians to apply ethical and regulatory principles with the help of an explicit RACI-style framework of accountability, regarding who is the Responsible, Accountable, Consulted, and Informed at every tier of authority.

To ensure traceability of data flows across EU migration management systems, independent monitoring bodies such as the European Data Protection Supervisor (EDPS) and EU-LISA, the European Union Agency for the Operational Management of Large-Scale IT Systems, are expected to conduct regular transparency audits,

issue seasonal accountability reports, and maintain secure audit trails [12]. These monitoring bodies currently oversee systems such as the EURODAC which captures fingerprints of asylum applicants to support the identification of individuals across the EU, EES which records the dates and biographical/biometric information at entry and exit of non-EU nationals at Schengen borders to monitor overstays [13] and ETIAS which is used pre-screen visa-exempt travellers prior to arrival into Schengen Zones to enhance border security and identify potential risks [14].

At the same time, stakeholder inclusion must be implemented as a structural component of stewardship. This means that human right NGOs and international organizations such as the UNHCR, are formally and actively involved in the processes of governance reviews whereby those who are most at risk of being affected by migration data systems are not ignored. In addition to compliance, this multi-layered stewardship model also encourages a rights-focused use of data, which maintains integrity, transparency, and sustainability through its lifecycle [15].

Simultaneously, algorithms used for decision making purposes are to be classified as a considerable risk as stated in the EU Artificial Intelligence Act [16]. In migration management, these systems are progressively underwriting risk scoring, profiling, and predictive analytics, which can greatly contribute to bias and discrimination. To combat against such risks, it should be mandated that independent bias audits are done before any system is deployed such that models are audited to be free, accurate, and diverse. Moreover, lifecycle performance monitoring of the system is necessary to identify and resolve any misalignments (data or ethical) of the system over time. These practices are embedded in line with the ACM Principle 1.4 [17], which is to be just and avoiding discrimination, aims to ensure that algorithmic tools in migration governance promote justice and equality instead of enshrining systemic bias.

Poor data governance is more than providing technical vulnerabilities. It is driving factor in the erosion of sustainable development. The principle of transparent and accountable institutions is focused by SDG 16, but the lack of transparency of surveillance systems, the inadequate recording of

DPIA processes, and the persistent lack of effective corrective mechanisms diminishes the population's trust in data governance systems [18]. Equally, SDG 10, which aims to minimize inequalities, is undermined when migrants and refugees become unevenly exposed to the dangers of technological systems. These gaps in governance have worsened digital inequality and given more privileged individuals greater safety leaving marginalized populations subject to exploitation risks.

Therefore, it is important to improve transparency, accountability, and human-centred governance to harmonize the practices of migration data with the objectives of sustainability. Governance of technological sustainability requires long-lasting legitimacy and fairness. It is under these progressions that data-driven migration systems can cultivate equity and institutional trust as opposed to continued harm and exclusion.

II. DATA PRIVACY AND PROTECTION

The ability to protect data and privacy forms the basis of organizational responsibility and ethical management of data. In the EU, the use of biometric and personal data in migration management has rapidly increased, with state and humanitarian actors becoming more actively involved as highlighted by the FRA [19]. This expansion of its use that has been frequently justified in terms of efficiency and security, raises significant ethical and legal challenges such as data rights, informed consent and accountability. Minorities, migrants, and refugees are among the most unprotected individuals in society. Moreover, research conducted by Human Rights Watch [20] has revealed that the vulnerable populations are exposed to unsafe and non-transparent data collection practices.

EU migration management systems increasingly rely on biometric technologies such as fingerprinting, facial recognition and iris scanning. These systems currently register and track individuals using this data [21]. The Eurodac and Schengen Information System II are designed to facilitate law enforcement, migration control, and the processing of asylum across EU member states. However, the sharing of this data with third parties/countries systems, e.g. the United States, is alarming in terms of privacy concerns and one's ownership over their own data. This can result in

lack of data protection in which informed and voluntary consent is compromised.

Studies have pointed out that migrants and refugees are usually obligated to give biometric information to get the necessary help, which may be food, accommodation, legal or identification services. They are often unaware of where their data will be used or who will access it. The practice eliminates any form of true choice and turns consent into coercion, thus risking GDPR Article 6 and 7 that considers the right of consent and extent of which the data can be used for processing. In times, when people are forced to choose between life and privacy, consent could be deemed ethically worthless. This type of circumstance poses a direct breach of the principle of respect to vulnerable people since they compel people to sell their primary rights to get instant relief. This situation is a severe indication of an ethical failure in the way data is obtained and managed, particularly where human dignity should be the primary concern. This highlights how ethical efforts may be disregarded when this is there is a lack of ethical frameworks in place.

Furthermore, organizations collecting such data may inadvertently allow government sectors to access or share this data unlawfully. For example, in the 2024 case in Greece [6], involving data protection authorities, biometric data collected from refugees for asylum was not disclosed regarding the potential data sharing with other systems or parties. This concern also applies to the proposed Enhanced Border Security Partnership. This type of data transfer is subject to violating many of the seven protection principles of the GDPR and the EU Charter of Fundamental Rights principles. Such cases highlight the importance of ethical standards in the collection, processing, and storage of personal data. Additionally, Frontex (the European Border and Coast Guard Agency) delegates data processing tasks relating to the sharing of information with non-EU states [22]. This partnership generally is outside the scope of GDPR oversight and illustrates a gap between the legality focus and everyday operational practices. This will similarly apply to the EU-US negotiations that are currently taking place through the EBSP.

The GDPR, which came into force in 2018, is still the strongest and most comprehensive data protection regime in the world. It states that

personal data must be processed lawfully, impartially, and in a transparent manner [2]. Under the GDPR, data collected from an individual must be used only when necessary and stored only for as long as required, with transparent processes clearly communicated to the individual.

The handling of sensitive data, such as biometrics or personal health information, is addressed in Articles 6 and 9 of the GDPR, which outline how such data should be processed within migration management systems. Moreover, Article 35 requires organisations to conduct a DPIA when processing sensitive information. However, despite this robust legal framework, enforcement is inconsistent, especially in cross-border contexts in multi-national and humanitarian situations. The power imbalance between migrants and data controllers complicates the true meaning of 'freely given consent'. Moreover, claims made under the pretext of national or public security which often used as excuses, weakens the responsibility of data controllers to protect personal data.

It is important that EU institutions and philanthropic supporters apply privacy centric approaches, where data protection is at the forefront, into any newly introduced data management systems. This means at every stage - from the collection to deletion stages. It is equally important that independent data protection officers be appointed, and transparent audit systems are put in place so that accountability is ensured from an early stage.

From a technical perspective, protecting data privacy requires secure storage and processes such as anonymization and encryption. Anonymization minimizes the dangers of exposing personal identifiable information (PII) and pseudonymization provides additional protection by substituting identifiers with artificial keys. The use of anonymization entirely is nearly impossible, especially in biometric systems as the data is unique for each subject. Therefore, systems must employ encryption, continuously log all data access and have special access privileges to a select few ensuring strong access controls.

Therefore, governance systems should adopt a multi-stakeholder oversight framework. This means, in addition to government and law enforcement agencies, independent, possibly third

party, data protection authorities and civil society organizations should also be included in the governance structure. This would enhance transparency and align more closely with the principles of accountability and proportionality set out under GDPR compliance standards. Privacy Impact Assessments (PIAs) are vital governance tools for evaluating risks and proactively identifying mitigation strategies before undertaking any large-scale data collection [23].

Right-based organizations should establish ethical review boards to assess data collection initiatives and determine whether they are necessary or are potentially harmful regarding privacy rights. Given the significant potential for harm, it is imperative these parties abide by the principle that they must "do no harm" to individuals, which adversely can result in discrimination or deportation.

Balancing data protection with national concerns such as border security remains a complex challenge in migration governance. Governments often justify significant data collection on the basis that it is required for managing migration flows, effectively identifying individuals, or preventing human trafficking and fraud. However, when data is retained or shared without the necessary required checks, this can ultimately result in a breach of privacy but also to a loss of public confidence. Therefore, it is important that a risk assessment approach is used when considering each data processing step [24].

Transparency plays an integral part in achieving this balance. Migrants and refugees should be informed at every stage of the process such as how their data is processed, stored, who may access it, and how long it will be retained. This can be done with the use of easily accessible and multilingual communication streams. Migrants should be able to challenge data misuse and request deletion under the GDPR's "right to be forgotten" [25] through clearly defined data retention policies and ability to access/modify data through applicable correction procedures. Future governance frameworks must also stress the importance of cross-border cooperation and oversight. Because data often flows across national borders, a well-coordinated governance framework is needed to ensure consistent application of privacy standards across all member states and partner countries (such as the US). It could be beneficial to establishing an EU

Migration Data Ethics Board that comprise of a legal department that actively works with the data developers, analysts and general society. The board could audit any large-scale migration databases or predictive technologies used for detection and issue seasonal transparency reports [26]. This independent oversight would lend legitimacy and increase public confidence in these institutions.

In summary, data privacy and protection are not only legal obligations but also need to consider the moral implications. The wrongful use of biometric data and of personal data threatens the very foundations of human rights regarding our privacy. Compliance with the GDPR, along with robust technical and transparent guidelines, is essential for institutional legitimacy and the protection of vulnerable groups. Aligning these practices with SDGs 10 and 16 reinforces the EU's commitment to reducing inequalities, promoting justice, and strengthening institutions. Ultimately, the integrity of data governance and protection in STS depends on recognizing privacy not as a privilege, but as a fundamental human right.

III. ETHICAL, LEGAL AND SOCIAL IMPLICATIONS

Ethical, legal and social aspects play a significant role in data driven systems. Biometric technology implementations have been rapidly adapted globally to ensure effective boarder clearance processes, in aims to detect fraud, improve efficiency and enforcement [27]. However, the record-breaking fine placed upon the Greek Ministry of Asylum and Migration [6] and the recent EU-US negotiations [8] on biometric data highlight the importance of both a comprehensive data protection and ethical impact assessment. Data-driven technologies which involve biometric and personal data of migrants within the European Union encompasses a plethora of social, cultural, and ethical beliefs and backgrounds. Therefore, it is fundamental that there is a commitment to ensure the safeguarding of human rights, values, and the dignity of vulnerable populations. From a sustainable perspective, the rapid expansion of these data solutions, raises concern regarding the long-term oversight and institutional integrity regarding accountability and promoting transparency.

From an ethical perspective, the unrestricted and

unregulated access to this data is of great concern. Recent research [28] shows that biometric data can reveal sensitive information about individuals such as ethnicity, gender, and possible underlying medical conditions that could threaten the right of privacy, autonomy, and may enable discrimination. There are a few ethical evaluation theories that can be highlighted when considering data driven technologies such as utilitarian ethics that is outcome focused, deontological ethics that is duty based, virtue-based ethics being focused on morals and Rawls' veil of ignorance which focusses on fairness and equality for all.

From a utilitarian perspective, biometric technologies have been used to provide efficiency through cost cutting (human workload), convenience, increased productivity by limiting the need for queues, increased mobility by making border crossing much more convenient, aid in fraud detection and greater border safety [27]. Thus, this approach would be considered ethical as it eliminates the potential human error or human biasness in identification for a greater sense of security.

Secondly, from a deontological perspective, there are several ethical concerns and moral obligations that should be upheld which includes the fundamental right to privacy and consent which is included in the GDPR [2] which regulates the use of any biometric data. Therefore, it is important for a DPIA to be carried out thereby identifying and proactively assessing and resolving any risks associated with data protection laws and human rights.

Biometric data does introduce additional risk if personal data is not correctly protected, is used without permission or if the participants cannot request deletion of such data. Migrants' who have had experiences with oppressive regimes have a distrust and concern in authority whereby there is a fear of sharing identity data which could be misused for persecution or repression for themselves and their families back home [29]. Political instability can further amplify this concern. In August 2021, the Taliban had seized extensive biometric databases placing thousands of vulnerable individuals at risk [30]. Additionally, Amnesty International [31] has reported that a lack of meaningful consent when it comes to the identification processes can escalate to the use of force where consent is often in the grey areas. The

implementation of such technologies is not primarily concerned with the actual welfare of its subjects (migrants, refugees) and prioritizes security and administration efficiency thereby opposing Kantian ethics and principles [32] as it undermines moral obligations.

Considering virtue ethics, one of the fundamental questions that should be considered is “What virtues does an institution embed into the design and operations of biometric systems?”. Fair treatment, trust, respect, empathy and transparency are expected in a virtuous institution.

In recent years, there has been numerous reports and media coverage regarding the systematic biasness of different groups of individuals. The result of such can be alarming as a large study has reported that there is a disproportionate high arrest search rate of African Americans based on decisions made by automatic facial recognition (AFR) software that is unregulated, has poor accuracy and lack of accountability [33]. In such cases where minority communities are disproportionately affected and can potentially exacerbate any existing discrimination thereby contradicting moral standards. AFR technologies, deployment of surveillance systems and predictive policing can result in fundamental freedoms and privacy being threatened. It is therefore important that institutions remain accountable, act responsibly, implement robust governance frameworks and there is public trust when it comes to the use of these technologies. Hence, virtue ethics prioritises the morals of such technologies and the character of implementing institutions.

Finally, with regards to Rawls’ veil of ignorance, this ethical theory requires factors such as ethnicity, gender and socio-economic background to be ignored with no group having any preferential treatment. However, the use of predictive technologies or surveillance technologies that systemically disadvantage certain populations as mentioned earlier, would be considered unethical in this theory. In addition, there are concerns raised regarding biometric data collection and application regarding senior citizens, unaccompanied children and people with disabilities, given their heightened vulnerabilities [34]. Consequently, it is important with the use of these technologies, institutions including border authorities, have the duty to uphold rights considering the circumstance with aims to protect the most vulnerable communities.

From the above ethical considerations, while the utilitarian framework these technologies provides a greater quantity of good regarding safety and efficiency, the virtue ethics, deontology and veil of ignorance cannot be ignored. It is integral that such technologies implemented from organisations integrate the security objectives with human rights legislation to ensure that the participants fundamental rights are upheld that is both ethical and sustainable. Ethical Impact Assessments (EIAs) should be carried out from the design to implementation stage in which each of the risk points raised should be considered with possible mitigation measures. It should also proactively find solutions to any ethical challenges that could arise such as bias or discrimination.

The legal frameworks of data collection are complex with regards to government goals and individual rights. One of the major concerns regarding biometric data is ensuring the collection, storage and processing complied with strict data protection requirements. As a result, both public and private organisations should be transparent regarding the data practices, implement strong security measures and ensure consent is received before data is collected [35].

As stated, the GDPR principles constitute part of the EU biometric law and requires that biometric data be processed only with explicit consent and control and requires that data controllers implement measures to prevent misuse (Article. 32 (2) of the GDPR [36]). A solution that has been provided to be privacy compliant when working with a biometric database is the use of algorithms that allow synthetic biometric data [37]. Failure to comply with these regulations can result in hefty fines or even sanctions, as imposed on the Greek case for the Kentauros and Hyperion [6]. This highlights the importance of DPIAs, moreover the lack of accountability, transparency and systematic weakness from a governance perspective on this ministry.

Another significant legal issue concerns the international transfer of biometric data for immigration management which requires international cooperation and sustainable legal frameworks which can be complicated by differing laws and policies [38]. This might be challenging from a design perspective of these technologies and should be overseen during DPIA stage to prevent any political misuse, ensure accountability and

transparency. Regulation (EU) 2016/399 and its most recent amendment Regulation (EU) 2024/1717 require that people be checked against criminal databases such as the Schengen Information System (SIS II) and Stolen and Lost Travel Documents (SLTD) but only on a risk-basis without systematic access [1]. However, the proposed Enhanced Border Security Partnership (EBSP) between the US and EU is risking EU laws being violated such as the GDPR and the EU-US Umbrella Agreement which regulate sharing of data with law enforcement while preventing continuous, automatic access.

The social implications of these technologies can reinforce social inclusion or exclusion. The use of biometric information for identification purposes at borders can aid in discriminatory racial profiling [39] or exacerbate existing social inequalities. Additional religious, sociocultural and personal factors must be considered, such as ethnic headscarves, face coverings, beards, headwear or interpersonal contact (such as not participating in photographs, being touched for fingerprint scanning), which could result in embarrassment or offence when dealing with technologies that are impractical. This can result in inequitable treatment or social exclusion of vulnerable groups and can create further social harm if an individual is unable to verify their data, participate in the biometric collection or even have hesitancy with the use of these technologies.

Additionally, the EU New Pact on Migration and Asylum has sparked strong criticism regarding the risks of human violations in which there are intensive preliminary data collection procedures, screenings and monitoring of asylum seekers, thereby stripping away their rights and could result in a negative perception of being framed as potential threats from a security/fraud perspective [40],[41]. This can result in negative public perception of these vulnerable communities, making social inclusion and access to opportunities more difficult. This can heighten fear and mistrust with both native populations and migrants, undermining national cohesion.

Furthermore, it is important to understand that these individuals could be hiding or felling from harm, whether that is inflicted upon from their government, gangs or even family in situations such as domestic violence, forced marriage or threats of honour killing. Biometric data can be

quite revealing providing information regarding underlying health conditions or even familial ties, and if exposed, can result in emotional, social harm or financial exclusion.

While data driven technologies relating to immigration provides significant growth in improving national security, streamlining delivery and aiding in administration, there are several ethical, legal and social implications that needs to be taken into consideration while ensuring that sustainable goals are met.

Ethical principles should guide both the development and implementation stages to ensure compliance with human rights legislation, respect for human dignity, and the provision of informed consent, as this directly contributes to SGD 10, which seeks to ensure inclusion and equal treatment. Striking the balance between the reaping the benefits and safeguarding these rights is complex. However ongoing engagement with lawmakers, developers, global experts and the wider community is essential in ensuring a successful and sustainable approach. In this regard, SGD 16, relies on institutions being transparent, accountable and fostering a global sense of trust and cooperation.

IV. DATA GOVERNANCE, ETHICS AND SUSTAINABILITY TO ADDRESS THE SGD(s)

Creating Fair and Responsible Biometric Migration Systems (SDGs 10 & 16)

EU authorities have started using more data especially fingerprints and facial scans to manage migration. This aims to improve security and reduce fraud at borders, however, presents big ethical and social questions, particularly for people in already vulnerable situations like migrants and refugees [27], [28].

Previous sections addressed the unclear rules, lack of consent, and technical mistakes that can result in violation of people's trust being broken in these systems. When this happens, it becomes more challenging to achieve the bigger goals of reducing inequality (SDG 10) and building trustworthy institutions (SDG 16) [1]. In this section we look at how to design biometric migration systems that are ethical, sustainable, and fair. This isn't just about following laws like GDPR or the EU AI Act but it's about truly respecting human rights, being transparent, and making sure migrant voices are

heard. Drawing on both regulations and ethical philosophies (like deontology or Rawls' veil of ignorance), this framework is meant to help and ensure data is handled fairly, keeps people safe, and builds public trust to the highest of standards.

Building ethical governance for biometric systems good governance is the backbone of any ethical migration system. Currently, the EU uses databases like Eurodac and the Schengen Information System, which is safeguarded by many national and EU-level laws [4]. Nevertheless, recent audits like the 2024 Greek data ruling show ongoing problems in risk assessments and sharing data safely across borders [6].

A comprehensive compliant policy:

A better approach includes several integrated layers that every system should meet, that align the legal requirements such as the GDPR, the EU AI Act, industry standards and human rights standards [2], [3]. This includes the collection, storing and processing of this information. Additionally including guidelines on:

Strong leadership and accountability: Dedicated teams oversee how data is handled from start to finish, keeping track of who does what at every level (such as the use of a RASCI). Public trust grows when people see themselves in the process.

Regular privacy and ethic reviews: It is important that high-risk activities get regular privacy and ethics reviews and updates. This ensures that the policy is evolving and complaint with any technological advancements and legal changes.

Input from everyone: Groups representing migrants, civil society, and bodies like the UNHCR should have a real say in how systems are managed and improved.

Technical Safety: Security measures like encryption and strict access controls are essential. Audits for fairness and biases in AI reduce chances of systemic discrimination [37]. This layered structure means data is managed responsibly at every stage, risks are reduced, and people can have more confidence in the system.

Turning principles into Real-World action for this approach to work, organizations need practical steps:

Balancing ethics and international utilitarian goals

It is fundamental that ethics is made central to data use by ensuring that people are prioritised. This can be achieved by:

Clear consent: Migrants deserve to understand why their data is collected and how it's used, not just written in fine print [5], [29]. Building trust with individuals empowers them to make informed decisions about their personal data.

Less Is More: Only collect data that is absolutely necessary (data minimization), with strict limits (purpose limitations) on reusing or sharing data [7]. This avoids the risk of any data breaches and privacy violations. Additional techniques for building privacy from "day one" is the use of techniques such as encryption and anonymization. This is a way to protect individual identities while being able to derive insights from this data. It is important that organizations proactively adapt to technological advancements to ensure that privacy rights of individuals are upheld.

No Discrimination: Systems need checks to avoid bias, regardless of background, gender, or status [33]. Applying Rawls' philosophy helps keep things fair [42]. This includes the use of AI models which requires regular testing and the use of diverse datasets for training and bias correction. It is important that there is human oversight of these technologies to ensure that any bias is identified and rectified effectively and prevent any social and legal ambiguities.

Transparency and Accountability: Easy-to-follow records and public reports help hold institutions to account [10] to identify any bias or privacy breaches.

Stakeholder Involvement: Regular input from NGOs, migrant representatives, and others ensures the system addresses real needs [41].

Easy Fixes and Feedback: Migrants should find it simple to correct mistakes, make complaints, or even ask for data to be deleted [9]. This mix of rules and real input links technical tools with human oversight, making systems safer and more trustworthy.

Incident Response Plans: Smart tools (such as dashboards) can catch problems like new biases or

risks, before they get out of hand [34] as an ongoing risk assessment, but an incident response plan and notification procedure should be developed once this is raised by the relevant department.

Independent Audits: External bodies must check that privacy and ethics guidelines are followed, and results should be public. This should be done periodically to mitigate any issues from an ongoing basis.

Internal Organization Training: Employee (such as border security) awareness programs and cybersecurity training can also ensure staff members fully understand compliance obligations and their role in data protection.

Robust Encryption of Databases: Use of advanced encryption should be utilized for data in transit (such as the EBS) or in idle data. An example is the use of blockchain technology which helps secure biometric data by providing multiple storage locations hence making it more difficult for unauthorized access or modification of data.

Supporting Global Goals for Equity and Justice: These frameworks do more than just tick boxes; they directly support the UN's SDG 10 (reducing inequality) and SDG 16 (building fair institutions). Proper oversight, ongoing audits, and genuine participation mean fewer inequalities for migrants and less risk of discrimination. Ensuring lawful, transparent, ethical, and sustainable data practices enhances the credibility of EU migration institutions and aligns operational procedures with international human rights standards [43].

Ultimately, incorporating ethics, good governance, and long-term thinking into EU migration systems isn't just the right thing to do, it's necessary for legal compliance and public trust. Case studies like Greece's show consequences of governance gaps, including fines, public distrust, and risks to vulnerable populations [5]. With this multi-layered approach: Everyone knows their responsibilities. Ethics guide every decision, from collecting consent to monitoring impact. Practical, accessible remedies are in place. Looking forward, there's a need to harmonize rules across countries and keep frameworks up to date as technology evolves. By embedding advanced encryption, strict access control, and staff training within ethical frameworks, the EU can make real progress toward

its development goals and guarantee that its migration systems are not just effective, but also fair, safe, trustworthy and sustainable.

V. CONCLUSION

To conclude, this paper reveals that even though the European Union have achieved advanced data encryption and protection systems such as the GDPR, the EU Charter of fundamental rights, and the AI act, there are still there many challenges that needs to be addressed for migration data governance. This report illustrates the extent that biometrics and personal information has been unlawfully and unethically scattered along the migration process. In different sections, it is observed that the vulnerability of these systems accounts to improper management of data. This refers to the inadequate enforcement, fragmented accountability and improper ethical communication. To achieve Sustainable Development Goals 10 and 16, a more ethical, rights based and sustainable approach is essential. This includes transparent governance, periodic audits, and capacity-building initiatives for policymakers and humanitarian actors to guarantee fairness. Moreover, migration system must be overlooked by ethical integrity, social responsibility and human rights to ensure the rightful and responsible use of migration data, thereby ensuring that EU migration governance upholds the principles of equality, openness, and sustainable development.

Future Research and Recommendations for Organizations

The future direction includes focusing on the advancements in ethical, technical and legal technologies or services in EU migration data governance to aim for more sustainable development. To enhance biometric data security and ensure compliance with legal standards, organizations should adopt the following practical measures:

- Create incident response plans: develop, implement and periodically access the functionality of the procedures to quickly address the biometric data security incidents.
- Provide Training to employee/faculties: Train the employees on regular basis to enhance the awareness about biometric data security.

- Advanced Data Encryption: Implement advance Encryption techniques so the data is not tampered with.
- Develop stronger comprehensive data protection policies: Create, maintain and continuously monitor the biometric data protection policy making it automated so it is updated and will show technological and regulatory advancements.

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